

This paper studies the effects of debt overhang on investment and payout policies using historical data from the Great Depression. Specifically, the authors use uncertainty about actual leverage due litigation over gold peg clauses in corporate bonds after the US went off the gold standard. Firms whose bonds had gold clauses were uncertain whether their leverage would be higher due to the sharp depreciation of the dollar against gold if the gold clauses were upheld in court. Consistent with theory, the authors find that firms exposed to this uncertainty reduced investment and increased payout to shareholders. The authors also find a reversal after the uncertainty was resolved by a Supreme Court ruling.

The paper follows a well-trodden path of effects of debt overhang but, in my view, adds two important pieces. First, novel identification: the debt overhang emerged because of uncertainty about leverage, which is exogenous as the decision to abandon the gold standard was arguably beyond the firms' control. Second, understanding the investment dynamics during the Great Depression by bringing new, disaggregated data to the table. Notably, the authors' bond data are not related to bank lending, which is usually the focus of Great Depression studies.

I have, however, a number of concerns that the authors should address:

1. Framing: The paper is closely related to firm-level uncertainty after Brexit studied by Hassan et al (2019, not cited). While a whole cottage industry has followed the firm-level approach pioneered by Hassan et al, I am not aware of a paper that focuses on an uncertainty shock to financial arrangements. Therefore, identifying a case of firm-level uncertainty about their financial contracts is a significant contribution to the literature. While the authors briefly point to Bloom (2009), the authors should highlight these firm-level differences in uncertainty much more and emphasize that these differences matter for aggregate outcomes during the Great Depression. Given the nature of the shock and the resulting litigation, I don't think the effects are due to "leverage risk" but to uncertainty about leverage. By reframing the paper and linking it to the literature on firm-level uncertainty, the paper would have a larger impact.

2. There are several issues with the empirical specification:
  - a. Why don't the authors use all data in the regression and estimate the treatment and the reversal in one regression on a balanced panel? This way the firm effects are consistently estimated.
  - b. There seems to be entry and exit, so it would be useful to know whether firms with gold clauses were more likely to exit. The answer is likely to be no given the results for profits. This would help the authors to the extent that it complements the profit non-result, as one would expect from an uncertainty shock. More generally, the authors should consider the effects of entry and exit in their calculations in table 11.
  - c. If available, I would like to understand whether the presence of the gold clause is related to the bond origination date. Specifically, I would expect gold clauses to be more likely to be present, the later the bond was issued (closer to 1933) since some countries had already left the gold standard. If true, the authors should also provide evidence that the timing of issuances does not affect their results (selection).
  - d. In table 6 the authors control for several balance sheet variables, some of which could be changing due to the gold clauses. The authors should present results that show whether gold clauses affected leverage and cash holdings as one response to uncertainty would be increasing cash and reducing leverage. This could be shown in table 6 and columns 2-8 could be relegated to the appendix.
  - e. Following up on the previous point, the authors could present more results on financing constraints. In addition to credit rating in table 5, showing whether the effect is larger for smaller firms, firms with low cash holdings, firms with higher book leverage would be standard in a paper like this.
  - f. I would also like to see some additional specification for the intensive margin: consider a dummy for high exposure at different cutoffs,  $d > 0.2$ ,  $d > 0.4$ .
  - g. The authors need to cluster standard errors on a time dimension. Given the number of years in the sample, clustering on the industry-year level is appropriate.
  - h. In table 11, I would also show a panel for firms with  $d=0$ .
3. When reframing the paper, please shorten the introduction. The current version is too long and meandering.

## References

Hassan, Tarek A., Stephan Hollander, Laurence van Lent, Ahmed Tahoun. 2019. "Firm-Level Political Risk: Measurement and Effects," *Quarterly Journal of Economics*, Volume 134, Issue 4, pp. 2135–2202.